

MADAPATI ABHINAV SAI

B.Tech — Computer Science & Engineering (AI & Machine Learning) | VIT-AP University | Amaravati, Andhra Pradesh, India
abhinavsaimadapati@gmail.com | +91-7981418181 | linkedin.com/in/madapati-abhinav-sai | github.com/abhinavsai2006 | abhinavsai.com

PROFESSIONAL SUMMARY

AI/ML-focused Computer Science undergraduate at VIT-AP University with hands-on experience designing, training, and deploying deep learning systems and production-grade full-stack applications. Fine-tuned BERT models achieving **86% test accuracy** across multi-domain NLP benchmarks; benchmarked three YOLO generations (v8/v10/v11) on a **1.55 GB geological dataset** across five destruction categories. Contributing as **Founding Software Engineer at Qaulium AI** — building product infrastructure spanning full-stack systems, authentication, database architecture, and CI/CD pipelines. Delivered **6+ end-to-end projects** across computer vision, NLP, and SaaS engineering using Python, PyTorch, React, and FastAPI. Seeking an AI/ML or software engineering internship to apply research depth and production engineering experience to real-world challenges.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, Bash

AI / Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, YOLO (v8/v10/v11), OpenCV, BERT (Fine-tuning), LSTM, Transformers, HuggingFace, Transfer Learning, Feature Engineering, Hyperparameter Tuning

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Data Preprocessing, Model Evaluation (F1, ROC-AUC, Confusion Matrix)

NLP: Text Classification, Sentiment Analysis, Sequence Modeling, Named Entity Recognition, Word Embeddings

Frontend: React, HTML5, CSS3, Tailwind CSS, Vite, Responsive Design

Backend / Databases: FastAPI, Node.js, Express, PostgreSQL, MySQL, MongoDB, Firebase, REST API Design, JWT Authentication

Tools & DevOps: Git, GitHub, Docker, Vercel, Cloudflare Workers, CI/CD Pipelines, Jupyter Notebook, VS Code, Linux

Concepts: Data Structures & Algorithms, Object-Oriented Programming, System Design, Agile, SDLC, Model Deployment, Software Prototyping

EXPERIENCE

Founding Software Engineer — Qaulium AI

Jan 2025 – Present | Remote, Early-Stage AI Startup

Full-Stack Engineering | TypeScript | React | Node.js | PostgreSQL | JWT | Vercel | CI/CD

- Architected and developed the full-stack product foundation from inception — including the **marketing landing page, admin dashboard, employee portal**, and internal engineering tooling.
- Implemented **JWT-based authentication** with role-based access control across admin and user-facing systems, ensuring secure session management throughout the platform.
- Designed **PostgreSQL database schemas** and REST API architecture to support scalable multi-role product flows; integrated email automation pipelines for transactional notifications and user onboarding.
- Owned end-to-end **deployment pipelines on Vercel with CI/CD integration**; maintained production environment stability across iterative shipping cycles in an Agile startup workflow.

PROJECTS

AI-Based Coal Destruction Classification System

[GitHub ↗](#)

Python | YOLOv8/YOLOv10/YOLOv11 | OpenCV | PyTorch | Computer Vision

- Engineered a multi-architecture object classification system, training and benchmarking **YOLOv8, YOLOv10, and YOLOv11** on a **1.55 GB geological dataset** spanning 5 standardized coal destruction categories.
- Built augmentation and preprocessing pipelines that improved model generalization; selected the optimal architecture using mAP50, precision, and recall as evaluation criteria.
- Implemented structured per-class evaluation tracking F1-score and confusion matrix across all three YOLO generations for systematic, reproducible comparison.

Multi-Domain Sentiment Analysis System

[GitHub ↗](#)

Python | BERT | LSTM | HuggingFace Transformers | Scikit-learn | NLP

- Designed and evaluated **15+ ML and deep learning classifiers** across **7 real-world NLP datasets**, systematically benchmarking architectures for multi-domain sentiment classification.
- Fine-tuned BERT and trained LSTM pipelines; achieved **86% test accuracy on the IMDB benchmark**, outperforming all baseline models including logistic regression, SVM, and naive Bayes.
- Generated production-quality per-domain evaluation reports covering accuracy, F1-score, confusion matrix, and ROC-AUC, enabling rigorous, data-driven model selection.

CricketConnect Pro

GitHub ↗

TypeScript | React | Node.js | PostgreSQL | Vercel | Cloudflare Workers

- Built and shipped a **production-deployed** cricket management platform supporting real-time venue bookings, tournament scheduling, and role-based multi-user access control.
- Secured multi-role authentication via JWT and optimized API response time using **Cloudflare Workers edge caching**, reducing server-side latency for end users.
- Maintained full deployment stability across **43 CI/CD iterations**, resolving cross-environment build failures and achieving complete mobile responsiveness.

LaundryHub — AI-Powered SaaS Platform

GitHub ↗

TypeScript | React | Firebase | Node.js | AI Integration

- Architected a **3-role SaaS platform** (Admin / Operator / Student) with real-time Firebase order tracking, live data synchronization, and push notification delivery across all user tiers.
- Integrated an AI-powered support chatbot that automated routine customer queries, reducing manual operator workload for high-frequency, repetitive request types.

Civic AI — Government Issue Reporting Platform

GitHub ↗

Python | FastAPI | React | NLP | REST APIs

- Developed an **NLP-powered complaint classification and auto-routing system** that dispatches civic issues to the correct government department, eliminating manual triage overhead.
- Built a FastAPI REST backend with priority-scoring and urgency-flagging logic; delivered a React dashboard enabling officials to filter, prioritize, and resolve incoming complaints.

EduVision-X

GitHub ↗

Python | React | Computer Vision | Deep Learning

- Architected an AI-enhanced educational platform integrating computer vision models to deliver **real-time, interactive feedback** to learners during active sessions.
- Optimized the model inference pipeline to reduce client-side latency, enabling responsive, low-delay performance for real-time interactive learning use cases.

EDUCATION

B.Tech — CSE (Artificial Intelligence & Machine Learning) VIT-AP University, Amaravati	2024 – 2028	CGPA: 7.5 / 10
Intermediate — Mathematics, Physics & Chemistry (MPC) Narayana Junior College	2022 – 2024	93%
Secondary School Certificate (SSC) Sri Chaitanya Techno School	2021 – 2022	93%

CERTIFICATIONS

- **Oracle Cloud Infrastructure 2025 AI Foundations Associate** — Oracle (2025)
- **Foundations of Project Management** — Google / Coursera
- **Data Analytics Virtual Job Simulation** — Deloitte Australia
- **Fundamentals of Java Programming**

ACHIEVEMENTS & ACTIVITIES

- **Top Contributor #1 — GDGoC VIT-AP Git & GitHub Open Source Challenge (Mar 2025):** Ranked highest among all participants in pull request contributions; formally recognised by GDG On Campus VIT-AP for outstanding open-source initiative during the event.
- **2nd Place — ElectroQuest Technical Competition,** IEI Student Chapter, VIT-AP University — competed in applied electronics and structured technical problem-solving.
- **Participant — VIT-AP Aura Hackathon:** Delivered a functional product prototype under time constraints as part of a cross-functional team.
- **AI/ML Team Member — Google Developer Groups on Campus (GDGoC), VIT-AP:** Active participant in peer-led AI/ML study groups, technical workshops, and community-driven development projects.